

Validation Stress Tests and Matched Smoothing Control

Final HalluGuard mechanism; MSE delta is percentage vs. no correction. Lower is better.

Experiment	Purpose	Final MSE Δ (%)	Matched smoothing Δ (%)	Gain vs. matched	Beats matched	Beats random	Takeaway
Boundary stress	Boundary jump repair	-2.486	-0.739	-1.746	16/16	16/16	Strong boundary-specific evidence
Slope break stress	Abrupt slope transition	-2.203	-0.719	-1.485	15/16	16/16	Robust to slope mismatch
Delayed level shift	Delayed horizontal offset	-2.161	-0.709	-1.452	16/16	14/16	Handles late level shift
High-frequency stress	Unsupported high-frequency perturbation	-2.895	-1.172	-1.723	16/16	13/16	Largest MSE reduction
Selected stress avg.	Four targeted stress families	-2.436	-0.835	-1.602	63/64	59/64	Not explained by equal-rate smoothing
Full stress suite	All six stress families	-2.509	-0.880	-1.629	95/96	89/96	General stress robustness

Matched smoothing control uses the same trigger/correction rate as the main method; beats columns count dataset-backbone-horizon configurations.